PCSPF Pancreatic Cancer Survival — Project Audit Documentation

Complete Technical Record, Methodology Audit, and Plain-Language Companion

Fundamentals of Data Science — CMSC 176

University of the Philippines Manila

Version 3.2 · May 26, 2026

PCSPF_Data_Science_Capstone.ipynb · 171 cells · 9 sections

878 patients · 20 features · seed 42

Table of Contents

1. Document Purpose and How to Read This Guide

This document is the formal audit record of the PCSPF capstone project. It documents every major process step, the technique used, parameters applied, outputs produced, and decisions made—including steps deliberately not taken. It reflects the notebook as executed through Section 9, including post-audit enhancements (overfitting analysis, metric clarifications, Cramér's V, model comparisons) and optimization sections (Why Random Forest?, Why K-Means?, enhanced supervised benchmark).

How to read: Each process entry contains (a) a technical description for instructors and reviewers, and (b) an "In simple terms" box for non-technical readers. Tables use wrapped text. Metrics match the executed notebook unless noted as approximate (~).

2. Executive Overview

Problem: Predict 1-year pancreatic cancer survival and discover patient subgroups using 20 preoperative clinical features from 878 patients. Methods: Random Forest (supervised, primary) and K-Means (unsupervised, k=3). The pipeline loads and cleans data (3 columns removed), explores distributions and correlations, documents preprocessing decisions, splits 80/20 stratified, clusters on scaled features, tunes Random Forest with GridSearchCV (class_weight=balanced), benchmarks alternative models, and synthesizes supervised and unsupervised findings in Section 8.

In simple terms

We studied hospital lab data to (1) estimate who is likely to live at least one year and (2) find natural groups of similar patients. We wrote down every step—including what we deliberately did not do—so the work can be verified, defended, and repeated.

Deliverable	Status	Location
Jupyter notebook (171 cells, 9 sections)	Complete	PCSPF_Data_Science_Capstone.ipynb
Figures (21 PNG, 300 DPI)	Generated on notebook run	figures/
Audit documentation PDF	v3.2	Documentations/PCSPF_Capstone_Project_Audit_Documentation.pdf
Notebook builder script	Available	scripts/assemble_notebook.py
PDF generator script	Available	scripts/generate_audit_documentation.py

3. End-to-End Pipeline Map

Phase	Notebook section	Key outputs
A	Setup	Imports, RANDOM_STATE=42, figures/ folder, plotting defaults
B	Load & clean	Excel drop 3 cols survival_label 878 × 21
C	EDA	Stats, 5 chart groups, correlation, class means, EDA summary
D	Preprocess	Types, nulls, outliers, normality, VIF, decisions table
E	Split & scale	702/176 stratified; StandardScaler on full X for K-Means only
F	Unsupervised	Method comparison elbow/silhouette K-Means k=3 PCA chi-square
G	Supervised	RF justification baseline GridSearchCV evaluation model comparison
H	Synthesis	RF vs cluster features, probability bridge, limitations, future work
I	Conclusion	Dynamic four-paragraph recap with runtime metrics

4. Audit Enhancements Register

The following cells were added after a full project audit to address gaps in justification, metric interpretation, and methodological transparency. No existing cells were deleted or reordered.

Location	Enhancement	What was added	Why it matters
Section 2	Pre-standardized data note	Documented that 18 continuous features in the released file are already z-scored (mean≈0, std≈1).	Clarifies why EDA ranges look standardized and why additional scaling is only emphasized for K-Means.
Section 6.5	PCA variance caveat	Added explicit note: PC1+PC2 = 30.8% variance; 69.2% of structure not visible in 2D plots.	Prevents over-interpretation of PCA overlap as proof clusters are meaningless.
Section 6.7	Cramér's V effect size	Computed Cramér's V ≈ 0.048 alongside chi-square p=0.2546.	Shows association is not only non-significant but also negligible in effect size.
Section 7.2	Three-objective RF tuning	Compared f1 (legacy grid), f1_macro (primary), recall_macro (stretch) with regularized hyperparameter grid.	Aligns tuning with macro F1 evaluation; documents legacy f1 row for before/after comparison.
Section 7.2	Threshold sweep	Exploratory probability threshold sweep on primary f1_macro model (default 0.50 retained).	Shows accuracy/recall trade-off without re-tuning on test labels.
Section 7.3	Train vs test overfitting table	Primary f1_macro RF: 85.8% train vs 61.93% test (23.8 pp gap) vs legacy f1 (100% / 71.02% / 29 pp).	Documents reduced memorization after macro-aligned tuning.
Section 7.3	Regularized RF comparison	Compared depth/leaf constraints showing accuracy-generalization trade-off.	Demonstrates regularization reduces overfit but does not reach clinical AUC threshold.
Section 7.3	Class-0 recall interpretation	Confusion matrix shows low recall for shorter-survival class (minority).	Clinical framing: model misses many high-risk patients despite balanced weights.
Section 7.5	Enhanced supervised model comparison	Benchmarked RF, LR, SVM RBF, GB with train/test gap, macro F1, AUC, CV macro F1.	Justifies RF retention on interpretability despite SVM's higher balanced metrics.
Section 6 (top)	Unsupervised method comparison	K-Means, GMM, Ward, DBSCAN compared on silhouette, chi-sq, cluster balance, assignment.	Justifies K-Means: best silhouette at k=3, full assignment, interpretable centroids.
Section 7 (top)	Why Random Forest?	Five-factor justification: non-linear interactions, Gini importance, multicollinearity, no scaling, course scope.	Pre-empts defense questions before baseline model section.

5. Techniques and Methods Glossary

Technique	What it is	How we used it	Key settings
Random Forest	Supervised ensemble of decision trees with bootstrap aggregation and random feature subsets.	Primary classifier for 1-year survival.	class_weight=balanced; primary f1_macro GridSearchCV; 100 trees, depth 5, leaf 4.
K-Means	Unsupervised partitioning minimizing within-cluster sum of squared distances.	Patient subgroup discovery.	k=3; StandardScaler features; n_init=10; seed 42.
Logistic Regression	Linear probabilistic classifier with log-odds link.	Benchmark in Section 7.5.	class_weight=balanced; scaled features.
SVM (RBF kernel)	Maximum-margin classifier in kernel-transformed space.	Benchmark; best macro F1 in comparison.	class_weight=balanced; probability=True.
Gradient Boosting	Sequential ensemble of shallow trees correcting prior errors.	Benchmark with native feature importance.	200 estimators; max_depth=3.
GMM / Ward / DBSCAN	Alternative clustering: probabilistic, hierarchical, density-based.	Unsupervised benchmarks in Section 6.	Compared at k=2,3,4; DBSCAN eps/ms grid.
StandardScaler	Z-score normalization: mean 0, variance 1 per feature.	K-Means input only.	Fit on full 878×20 matrix (unsupervised).
PCA	Orthogonal linear projection capturing maximum variance.	2D visualization of clusters.	PC1 20.3% + PC2 10.5% = 30.8% cumulative.
GridSearchCV	Exhaustive hyperparameter search with cross-validation.	Tune Random Forest.	Primary: 135 combos, scoring=f1_macro; legacy f1 row: 108 combos (reference).
Cramér's V	Effect size for categorical association (0 = none, 1 = perfect).	Cluster vs survival after chi-square.	V ≈ 0.048 — negligible.
VIF	Variance inflation factor for multicollinearity.	Section 4.6 audit.	Elevated on NLR, PLR, SII, CRP/ALB; none dropped.

6. Repository and File Inventory

File / path	Purpose
PCSPF_Data_Science_Capstone.ipynb	Main analysis notebook (all sections, audit cells, optimizations)
Dataset/PCSPF-Pancreatic Cancer Survival based on Preoperative Features.xlsx	Source data: 878 patients, 24 columns raw
figures/*.png	21 exported visualizations at 300 DPI
scripts/audit_content.py	Structured text content for this PDF
scripts/generate_audit_documentation.py	Regenerates this PDF document
scripts/assemble_notebook.py	Programmatic notebook builder
requirements.txt	Python package dependencies
Documentations/fonts/	Poppins font files for PDF typography

7. Complete Process Documentation (Sections 1–9)

The following subsections mirror the Jupyter notebook. Process IDs match notebook subsection numbers. Audit additions are marked (audit) in the process name.

SECTION 1: Project Setup and Imports

Established the computational environment, reproducibility controls, and plotting standards before any data were loaded.

In simple terms
We set up the software toolbox and fixed random seeds so every chart and model result can be recreated exactly the same way later.

Process 1.1: Library imports

Process ID	1.1
Step name	Library imports
Technique / method	Python scientific stack
What we did (detailed)	Imported pandas, numpy, matplotlib, seaborn, scikit-learn (RF, KMeans, PCA, StandardScaler, train_test_split, GridSearchCV, StratifiedKFold, metrics, silhouette_score), scipy (chi2, normaltest), statsmodels (VIF). SMOTE imported but not applied.
Parameters / settings	N/A
Output / artifact	All dependencies loaded; warnings suppressed.
Decision / rationale	SMOTE documented as rejected alternative.

In simple terms
We loaded programs for tables, graphs, and machine learning.

Process 1.2: Reproducibility

Process ID	1.2
Step name	Reproducibility

Technique / method	Fixed random seed
What we did (detailed)	RANDOM_STATE=42 on split, KMeans, RF, StratifiedKFold, PCA. OPTIMAL_K=3 documented.
Parameters / settings	random_state=42
Output / artifact	Identical splits and fits on re-run.
Decision / rationale	Mandatory for academic reproducibility.

In simple terms

We fixed the randomness so results do not change every run.

Process 1.3: Visualization defaults

Process ID	1.3
Step name	Visualization defaults
Technique / method	matplotlib + seaborn
What we did (detailed)	Style seaborn-v0_8-whitegrid; colorblind palette; figsize defaults; savefig dpi=300, bbox_inches='tight'; figures/ directory created.
Parameters / settings	dpi=300
Output / artifact	Publication-quality exports.
Decision / rationale	Consistent visuals for report and defense.

In simple terms

We chose clear, high-resolution saved images.

SECTION 2: Data Loading and Initial Inspection

Loaded Excel data, audited structure, removed unusable columns, renamed target.

In simple terms
We opened the patient spreadsheet, checked it, and prepared a clean table for analysis.

Process 2.1: Load Excel

Process ID	2.1
Step name	Load Excel
Technique / method	pd.read_excel
What we did (detailed)	Read PCSPF workbook. Verified shape (878, 24).
Parameters / settings	openpyxl
Output / artifact	DataFrame 878×24.
Decision / rationale	Single authoritative source file.

In simple terms
We imported 878 patient rows from Excel.

Process 2.2: Structural inspection

Process ID	2.2
Step name	Structural inspection
Technique / method	Descriptive audit
What we did (detailed)	Columns, dtypes, head/tail, null counts, duplicates.
Parameters / settings	N/A
Output / artifact	878 nulls each in Predict label columns; 0 duplicates.

Decision / rationale	Predict columns unusable; ID all 'xxx'.

In simple terms
We listed columns, types, and missing values.

Process 2.3: Target distribution

Process ID	2.3
Step name	Target distribution
Technique / method	value_counts
What we did (detailed)	Class 1: 604 (68.8%); Class 0: 274 (31.2%); ratio 2.20:1.
Parameters / settings	N/A
Output / artifact	Majority baseline = 68.8% accuracy.
Decision / rationale	Drives class_weight and metric choice.

In simple terms
About 7 in 10 patients survived at least one year.

Process 2.4: Column removal

Process ID	2.4
Step name	Column removal
Technique / method	drop + rename
What we did (detailed)	Dropped ID, Predict label 1, Predict label 0. Renamed target to survival_label.
Parameters / settings	DROP_COLS
Output / artifact	Clean shape (878, 21) = 20 features + target.
Decision / rationale	Asserted expected dimensions.

In simple terms

We removed useless columns and renamed the outcome.

Process 2.5: Pre-standardized note (audit)

Process ID	2.5
Step name	Pre-standardized note (audit)
Technique / method	Descriptive check
What we did (detailed)	Verified continuous features have mean≈0, std≈1 in released file.
Parameters / settings	N/A
Output / artifact	Markdown + code audit cell.
Decision / rationale	Explains EDA scale; scaling still applied for K-Means.

In simple terms

We noted the file already used z-scores for lab columns.

SECTION 3: Exploratory Data Analysis (EDA)

Characterized distributions, class balance, correlations, and survival-group differences.

In simple terms

We explored the data with charts and tables before building models.

Process 3.1: Summary statistics

Process ID	3.1
Step name	Summary statistics
Technique / method	describe()
What we did (detailed)	Count, mean, std, min, quartiles, max for all 20 features.
Parameters / settings	Transposed table
Output / artifact	Displayed in notebook.
Decision / rationale	Confirmed z-score scale in released data.

In simple terms

We calculated average and spread of each measurement.

Process 3.2: Target bar chart

Process ID	3.2
Step name	Target bar chart
Technique / method	matplotlib bar
What we did (detailed)	Class counts with percent labels target_distribution.png.
Parameters / settings	dpi=300
Output / artifact	Imbalance ratio 2.20:1 printed.

Decision / rationale	Foreshadows class_weight strategy.

In simple terms
We charted survivors vs non-survivors.

Process 3.3: Feature histograms

Process ID	3.3
Step name	Feature histograms
Technique / method	histplot + KDE
What we did (detailed)	4×5 grid for 18 continuous features feature_distributions.png.
Parameters / settings	30 bins
Output / artifact	Right-skew on CEA, CRP, NLR, bilirubin.
Decision / rationale	Informs normality testing in Section 4.

In simple terms
We drew distribution shapes for each lab.

Process 3.4: Survival boxplots

Process ID	3.4
Step name	Survival boxplots
Technique / method	boxplot by class
What we did (detailed)	Grouped by survival_label boxplots_by_survival.png.
Parameters / settings	colorblind palette
Output / artifact	Modest separation on CA19-9, inflammation.
Decision / rationale	Weak univariate signal motivates multivariate models.

In simple terms
We compared labs between outcome groups.

Process 3.5: Correlation heatmap

Process ID	3.5
Step name	Correlation heatmap
Technique / method	seaborn heatmap
What we did (detailed)	Pearson matrix, upper triangle masked correlation_heatmap.png.
Parameters / settings	Annotated r values
Output / artifact	All |r| with target < 0.15.
Decision / rationale	Strong inter-feature collinearity among ratios.

In simple terms
We mapped which variables move together.

Process 3.6: Multicollinearity narrative

Process ID	3.6
Step name	Multicollinearity narrative
Technique / method	Markdown
What we did (detailed)	Documented NLR, PLR, SII, CRP/ALB formulas.
Parameters / settings	N/A
Output / artifact	Conceptual link to VIF in Section 4.
Decision / rationale	Prepares defense on retaining correlated features.

In simple terms
We explained why some columns overlap clinically.

Process 3.7: Class-conditional means

Process ID	3.7
Step name	Class-conditional means
Technique / method	groupby + bar chart
What we did (detailed)	Mean diff class 0 vs 1; top 10 class_conditional_means.png.
Parameters / settings	Sorted by |diff|
Output / artifact	Largest gaps: CA19-9, Prealbumin, Neutrocyte, Abdominal Pain.
Decision / rationale	Quantifies weak univariate separation.

In simple terms
We measured which labs differ most by outcome.

Process 3.8: EDA summary

Process ID	3.8
Step name	EDA summary
Technique / method	Synthesis markdown
What we did (detailed)	Four themes: distributions, correlations, class gaps, modeling implications.
Parameters / settings	N/A
Output / artifact	Bridges EDA to preprocessing.
Decision / rationale	Single narrative checkpoint.

In simple terms
We summarized findings before modeling.

SECTION 4: Data Preprocessing

Documented every data decision including deliberate non-actions, with clinical justification.

In simple terms
We recorded every preparation choice—and what we chose NOT to do and why.

Process 4.1: Data types

Process ID	4.1
Step name	Data types
Technique / method	dtype audit
What we did (detailed)	All numeric; binary cols {0,1}.
Parameters / settings	N/A
Output / artifact	No casting needed.
Decision / rationale	Confirmed in markdown.

In simple terms
We checked column types.

Process 4.2: Missing values

Process ID	4.2
Step name	Missing values
Technique / method	isnull audit
What we did (detailed)	Triple check — all zero.
Parameters / settings	N/A
Output / artifact	No imputation.

Decision / rationale	Three verification methods.

In simple terms

We confirmed no empty cells.

Process 4.3: Duplicates

Process ID	4.3
Step name	Duplicates
Technique / method	duplicated()
What we did (detailed)	0 exact duplicates.
Parameters / settings	N/A
Output / artifact	All rows retained.
Decision / rationale	Documented.

In simple terms

No duplicate patients.

Process 4.4: Outliers

Process ID	4.4
Step name	Outliers
Technique / method	IQR 1.5× rule
What we did (detailed)	Counts per feature; z-score boxplot outlier_boxplots.png. Highest: CEA 10.4%; CA19-9 = 0% (wide IQR).
Parameters / settings	1.5×IQR
Output / artifact	Decision: RETAIN all outliers.
Decision / rationale	Clinical extremes are informative.

In simple terms

We counted extremes but kept them.

Process 4.5: Skewness & normality

Process ID	4.5
Step name	Skewness & normality
Technique / method	normaltest
What we did (detailed)	Skew/kurtosis table; D'Agostino-Pearson per feature.
Parameters / settings	alpha=0.05
Output / artifact	Most non-normal; CEA/PLR/NLR most skewed; CA19-9 moderate; no transform.
Decision / rationale	RF uses ranks; K-Means uses scaler.

In simple terms

We tested bell-curve assumptions.

Process 4.6: VIF

Process ID	4.6
Step name	VIF
Technique / method	variance_inflation_factor
What we did (detailed)	VIF for all 20 features; sorted interpretation table.
Parameters / settings	statsmodels
Output / artifact	High VIF on ratio features.
Decision / rationale	Decision: RETAIN all features.

In simple terms

We measured overlap but kept clinical indices.

Process 4.7: Feature engineering

Process ID	4.7
Step name	Feature engineering
Technique / method	None
What we did (detailed)	Existing ratios sufficient; no binning or interactions.
Parameters / settings	N/A
Output / artifact	Markdown only.
Decision / rationale	Avoid arbitrary information loss.

In simple terms

No new columns created.

Process 4.8: Feature selection

Process ID	4.8
Step name	Feature selection
Technique / method	Retain all 20
What we did (detailed)	Same set for RF and K-Means; importance post-hoc.
Parameters / settings	N/A
Output / artifact	Consistent pipelines.
Decision / rationale	Same inputs both analyses.

In simple terms

All 20 features used.

Process 4.9: Class imbalance

Process ID	4.9
Step name	Class imbalance
Technique / method	class_weight
What we did (detailed)	Documented weights; rejected SMOTE.
Parameters / settings	balanced
Output / artifact	F1, AUC primary metrics.
Decision / rationale	Real patients only.

In simple terms

Model weights minority class more.

Process 4.10: Summary table

Process ID	4.10
Step name	Summary table
Technique / method	matplotlib table
What we did (detailed)	11-row decision register preprocessing_summary.png.
Parameters / settings	N/A
Output / artifact	Master audit reference.
Decision / rationale	Single consolidated table.

In simple terms

One table of all prep choices.

SECTION 5: Train-Test Split and Scaling

Separated X and y, stratified 80/20 split, scaling for K-Means only.

In simple terms
We divided data for fair testing and scaled only for grouping.

Process 5.1: Feature-target split

Process ID	5.1
Step name	Feature-target split
Technique / method	X, y
What we did (detailed)	X: 878×20; y: survival_label.
Parameters / settings	N/A
Output / artifact	Shapes printed.
Decision / rationale	Supervised foundation.

In simple terms
Separated inputs from outcome.

Process 5.2: Train-test split

Process ID	5.2
Step name	Train-test split
Technique / method	train_test_split
What we did (detailed)	80/20 stratified 702 train, 176 test.
Parameters / settings	seed=42
Output / artifact	Class proportions preserved.

Decision / rationale	Holdout for RF.

In simple terms

20% hidden for final test.

Process 5.3: StandardScaler

Process ID	5.3
Step name	StandardScaler
Technique / method	fit on full X
What we did (detailed)	Before/after stats; boxplot scaling_before_after.png.
Parameters / settings	StandardScaler()
Output / artifact	X_scaled for K-Means.
Decision / rationale	RF uses unscaled train/test.

In simple terms

Scaled labs for fair clustering distances.

SECTION 6: Unsupervised Learning — K-Means Clustering

Method comparison (K-Means, GMM, Ward, DBSCAN), model selection, clustering, profiling, and survival overlay without using target during fit.

In simple terms

We compared grouping methods, chose K-Means, found three patient types, then checked survival.

Process 6.0: Method comparison (audit)

Process ID	6.0
Step name	Method comparison (audit)
Technique / method	Multi-algorithm benchmark
What we did (detailed)	K-Means, GMM, Ward (k=2–4); DBSCAN (3 configs). Metrics: silhouette, chi-sq p, Cramér's V, survival spread, smallest cluster, assignment %.
Parameters / settings	Same X_scaled, seed 42
Output / artifact	unsupervised_model_comparison.png.
Decision / rationale	K-Means selected: best silhouette at k=3, 100% assignment, centroids.

In simple terms

We compared four grouping methods before choosing K-Means.

Process 6.1: Elbow method

Process ID	6.1
Step name	Elbow method
Technique / method	KMeans inertia
What we did (detailed)	k=2..10 on X_scaled elbow_method.png.
Parameters / settings	n_init=10
Output / artifact	Elbow region k=2–4.

Decision / rationale	Supports k choice.

In simple terms
Found where extra groups help less.

Process 6.2: Silhouette analysis

Process ID	6.2
Step name	Silhouette analysis
Technique / method	silhouette_score
What we did (detailed)	Mean silhouette per k silhouette_scores.png.
Parameters / settings	k=2..10
Output / artifact	Max at k=2 (~0.206).
Decision / rationale	Compared with elbow.

In simple terms
Scored cluster tightness.

Process 6.3: Choose k=3

Process ID	6.3
Step name	Choose k=3
Technique / method	Structured justification
What we did (detailed)	k=3 for three clinical phenotypes despite k=2 silhouette peak.
Parameters / settings	OPTIMAL_K=3
Output / artifact	Documented trade-off.
Decision / rationale	Clinical interpretability.

In simple terms
We picked three patient types.

Process 6.4: Final K-Means

Process ID	6.4
Step name	Final K-Means
Technique / method	fit_predict
What we did (detailed)	Labels for 878 patients; cluster sizes printed.
Parameters / settings	seed=42
Output / artifact	df['cluster'] assigned.
Decision / rationale	Target not used in fit.

In simple terms
Every patient got a group.

Process 6.5: PCA visualization

Process ID	6.5
Step name	PCA visualization
Technique / method	PCA n=2
What we did (detailed)	Variance ratios; scatter pca_clusters.png.
Parameters / settings	PCA
Output / artifact	PC1+PC2 = 30.8%.
Decision / rationale	PCA variance caveat added (audit).

In simple terms
2D picture of groups.

Process 6.6: Cluster profiling

Process ID	6.6
Step name	Cluster profiling
Technique / method	groupby + heatmap
What we did (detailed)	Mean/std tables; z-scored heatmap cluster_profiles_heatmap.png.
Parameters / settings	N/A
Output / artifact	Three clinical phenotypes.
Decision / rationale	Low/mid/high inflammation profiles.

In simple terms

Described each group medically.

Process 6.7: Survival overlay

Process ID	6.7
Step name	Survival overlay
Technique / method	chi-square + Cramér's V
What we did (detailed)	Crosstab; survival %; bar chart; chi2 p=0.2546; V≈0.048.
Parameters / settings	chi-square
Output / artifact	cluster_survival_overlay.png.
Decision / rationale	Not significant; negligible effect.

In simple terms

Groups don't strongly predict survival.

Process 6.8: PCA by survival

Process ID	6.8
Step name	PCA by survival
Technique / method	scatter by label
What we did (detailed)	Same PCA colored by true survival pca_survival_labels.png.
Parameters / settings	N/A
Output / artifact	Partial alignment with clusters.
Decision / rationale	Bridge to supervised outcome.

In simple terms

Overlaid survival on 2D map.

SECTION 7: Supervised Learning — Random Forest Classifier

RF justification, baseline, GridSearchCV, full evaluation with overfitting audit, regularization comparison, and multi-model benchmark.

In simple terms

We explained why RF, tuned it, measured overfitting, and compared to other classifiers.

Process 7.0: Why Random Forest? (audit)

Process ID	7.0
Step name	Why Random Forest? (audit)
Technique / method	Five-factor markdown
What we did (detailed)	Non-linear interactions, Gini importance, multicollinearity robustness, no scaling, course scope.
Parameters / settings	N/A
Output / artifact	Pre-7.1 justification.
Decision / rationale	Aligns with synthesis requirements.

In simple terms

We wrote why RF fits this project.

Process 7.1: Baseline RF

Process ID	7.1
Step name	Baseline RF
Technique / method	RandomForestClassifier
What we did (detailed)	100 trees, balanced; fit X_train.
Parameters / settings	n_estimators=100
Output / artifact	Acc 68.18% on test.

Decision / rationale	Benchmark before tuning.

In simple terms
First draft model.

Process 7.2: GridSearchCV

Process ID	7.2
Step name	GridSearchCV
Technique / method	Hyperparameter search
What we did (detailed)	Three objectives: f1 (legacy), f1_macro (primary), recall_macro (stretch).
Parameters / settings	Primary best: 100 trees, depth 5, leaf 4
Output / artifact	CV macro F1 0.579 (primary).
Decision / rationale	Legacy f1 row kept for comparison.

In simple terms
Aligned tuning with macro F1 evaluation.

Process 7.3: Test evaluation

Process ID	7.3
Step name	Test evaluation
Technique / method	Full metric suite
What we did (detailed)	Primary f1_macro RF: 61.93% acc; macro F1 0.550; AUC 0.563; CM; ROC; classification report.
Parameters / settings	176-patient holdout
Output / artifact	4 figures saved.
Decision / rationale	Class-0 recall 36.4% (20/55) documented.

In simple terms

Scored on unseen patients.

Process 7.3a: Overfitting assessment (audit)

Process ID	7.3a
Step name	Overfitting assessment (audit)
Technique / method	Train vs test table
What we did (detailed)	Primary: 85.8% train vs 61.93% test (23.8 pp gap). Legacy f1: 100% / 71.02% / 29 pp.
Parameters / settings	N/A
Output / artifact	Markdown interpretation.
Decision / rationale	Reduced but present overfit.

In simple terms

Memorization reduced vs legacy tuning.

Process 7.3b: Regularization comparison (audit)

Process ID	7.3b
Step name	Regularization comparison (audit)
Technique / method	Constrained RF configs
What we did (detailed)	Shallower trees / larger leaves reduce gap at accuracy cost.
Parameters / settings	N/A
Output / artifact	Trade-off table.
Decision / rationale	No config reaches AUC 0.7.

In simple terms

Stricter trees generalize better but score lower.

Process 7.3c: Class-0 recall (audit)

Process ID	7.3c
Step name	Class-0 recall (audit)
Technique / method	Confusion matrix narrative
What we did (detailed)	Low recall for shorter-survival class despite balanced weights.
Parameters / settings	N/A
Output / artifact	Clinical limitation stated.
Decision / rationale	Many high-risk patients missed.

In simple terms

Model under-detects worst outcomes.

Process 7.4: Feature importance

Process ID	7.4
Step name	Feature importance
Technique / method	Gini importances
What we did (detailed)	Bar chart all 20 features feature_importance.png.
Parameters / settings	Gini
Output / artifact	Top: CEA, Prealbumin, CA19-9.
Decision / rationale	Feeds Section 8 synthesis.

In simple terms

Ranked which labs matter most.

Process 7.5: Model comparison (audit)

Process ID	7.5
Step name	Model comparison (audit)
Technique / method	RF vs LR vs SVM vs GB
What we did (detailed)	Same split/seed; train acc, test acc, overfit gap, macro F1, AUC, CV macro F1.
Parameters / settings	Pipeline CV for scaled models
Output / artifact	supervised_model_comparison.png.
Decision / rationale	All AUC < 0.63; RF kept for importance.

In simple terms

Compared four prediction methods.

SECTION 8: Synthesis and Bridging Analysis

Connected RF importances, cluster structure, predicted probabilities, limitations, future work.

In simple terms
We compared predictions with groups to see if both analyses agree.

Process 8.1: Importance vs differentiation

Process ID	8.1
Step name	Importance vs differentiation
Technique / method	Rank comparison
What we did (detailed)	RF rank vs cluster mean-range rank synthesis_comparison.png.
Parameters / settings	N/A
Output / artifact	Top-5 overlap: CRP/ALB only; partial convergence.
Decision / rationale	Different questions, related biology.

In simple terms
Compared survival predictors vs phenotype separators.

Process 8.2: Probability by cluster

Process ID	8.2
Step name	Probability by cluster
Technique / method	predict_proba
What we did (detailed)	Mean predicted vs actual survival rate per cluster.
Parameters / settings	tuned RF on full X
Output / artifact	Bridge table.

Decision / rationale	Calibration across subgroups.

In simple terms
Compared model confidence per group.

Process 8.3: Clinical synthesis

Process ID	8.3
Step name	Clinical synthesis
Technique / method	Dynamic narrative
What we did (detailed)	High-risk cluster 2 vs low-risk cluster 0; convergence paragraph.
Parameters / settings	runtime vars
Output / artifact	Rendered markdown.
Decision / rationale	Medical story.

In simple terms
Wrote connected clinical interpretation.

Process 8.4: Limitations

Process ID	8.4
Step name	Limitations
Technique / method	10 items
What we did (detailed)	Imbalance, weak signal, overfitting, no external validation, K-Means assumptions, etc.
Parameters / settings	N/A
Output / artifact	Numbered list.
Decision / rationale	Honest scope.

> **In simple terms**
> Listed what study cannot claim.

Process 8.5: Future work

Process ID	8.5
Step name	Future work
Technique / method	10 items
What we did (detailed)	XGBoost, SHAP, Cox models, GMM/DBSCAN validation, external cohorts, staging features.
Parameters / settings	N/A
Output / artifact	Research directions.
Decision / rationale	Beyond course scope.

> **In simple terms**
> Listed next research steps.

SECTION 9: Conclusion

Four-paragraph recap with metrics from runtime variables.

In simple terms

The ending auto-fills real numbers from the executed notebook.

Process 9.1: Dynamic conclusion

Process ID	9.1
Step name	Dynamic conclusion
Technique / method	Markdown f-string
What we did (detailed)	Objective, supervised results, unsupervised results, bridging insight.
Parameters / settings	tuned_acc, auc, macro_f1, clusters, chi2 p
Output / artifact	KEY RESULTS block + rendered text.
Decision / rationale	Avoids stale hard-coded metrics.

In simple terms

Summary always matches latest run.

8. Preprocessing Decisions Register

Step	Action	Justification
Drop ID, Predict labels	Removed	ID anonymized; predict columns 100% NaN
Missing values	No imputation	Zero nulls after column drop
Duplicates	None removed	0 duplicate rows
Outliers	Retained	IQR documented; highest CEA 10.4%; CA19-9 = 0% (wide IQR)
Normality transform	Not applied	RF rank-invariant; scaler for K-Means only
Multicollinearity (VIF)	Retain all features	RF subsampling; ratio features clinically validated
Feature engineering	None added	NLR, PLR, SII, CRP/ALB already in dataset
Feature selection	All 20 retained	Post-hoc RF importance; same set for K-Means
SMOTE	Not used	class_weight='balanced' preferred for clinical defensibility
RF scaling	None	Tree splits are scale-invariant
K-Means scaling	StandardScaler on full X	Euclidean distance requires comparable scales
Train-test split	80/20 stratified	702 train / 176 test; seed 42

9. Visualizations Register (21 Figures)

#	Filename	Sec.	Description
1	target_distribution.png	3.2	Bar chart of class 0 vs class 1 with count and percent labels
2	feature_distributions.png	3.3	4×5 grid of histograms + KDE for 18 continuous features
3	boxplots_by_survival.png	3.4	Boxplots of each continuous feature by survival label
4	correlation_heatmap.png	3.5	Lower-triangle Pearson correlation heatmap (20 features + target)
5	class_conditional_means.png	3.7	Horizontal bar chart of top 10 mean differences by class
6	outlier_boxplots.png	4.4	Per-feature z-score boxplots for outlier comparison
7	preprocessing_summary.png	4.10	Table figure of all preprocessing decisions
8	scaling_before_after.png	5.3	Side-by-side boxplots before/after StandardScaler
9	unsupervised_model_comparison.png	6.0	K-Means vs GMM vs Ward at k=3 (silhouette, chi-sq p, spread)
10	elbow_method.png	6.1	Line plot of inertia vs k (2–10) with elbow marker
11	silhouette_scores.png	6.2	Line plot of mean silhouette vs k
12	pca_clusters.png	6.5	2D PCA scatter colored by cluster + centroids
13	cluster_profiles_heatmap.png	6.6	Heatmap of z-scored mean features per cluster
14	cluster_survival_overlay.png	6.7	Grouped bar chart of survival counts per cluster
15	pca_survival_labels.png	6.8	2D PCA scatter colored by true survival
16	confusion_matrix.png	7.3	Seaborn heatmap of tuned RF predictions
17	roc_curve.png	7.3	ROC with AUC annotation and diagonal reference
18	classification_report.png	7.3	Tabular precision/recall/F1 figure
19	feature_importance.png	7.4	Horizontal bar chart of all 20 Gini importances
20	supervised_model_comparison.png	7.5	RF vs LR vs SVM vs GB — accuracy, macro F1, AUC
21	synthesis_comparison.png	8.1	Side-by-side top-10 RF vs cluster-differentiation

10. Quantitative Results Reference

10.1 Dataset and split

Metric	Value
Patients	878
Features (predictors)	20
Target	survival_label (1 = survived ≥ 1 year)
Train / test	702 / 176 (80/20 stratified)
Class 1 / Class 0	604 (68.8%) / 274 (31.2%)
Imbalance ratio	2.20 : 1
Majority baseline accuracy	68.8%

10.2 Supervised learning (Random Forest)

Metric or item	Value	Notes
Untuned RF test accuracy	68.18%	100 trees, before grid search
Tuned RF test accuracy	61.93%	-6.87 pp vs 68.8% baseline
Tuned RF train accuracy	85.8%	Overfitting indicator
Train–test accuracy gap	23.8 pp	Documented in audit Section 7.3
Macro F1 (test)	0.550	Average of both classes
AUC-ROC (test)	0.563	Below 0.7 clinical utility threshold
CV F1 binary (tuning)	0.809 (legacy f1 row)	GridSearch scoring metric
CV F1 macro (comparison)	0.579	Fair cross-model benchmark
Best n_estimators	100	GridSearch winner
Best max_depth	5	GridSearch winner
Top 5 features	CA19-9, CEA, Prealbumin, CRP/ALB, BMI	Gini importance

10.3 Unsupervised learning (K-Means, k=3)

Cluster	N	% cohort	Survival rate	Clinical profile (summary)
0	519	59.1%	70.1%	Lower inflammation; higher prealbumin/ALB
1	204	23.2%	69.6%	Elevated bilirubin, NLR, SII; lower prealbumin

Cluster	N	% cohort	Survival rate	Clinical profile (summary)
2	155	17.7%	63.2%	Highest CRP/CRP/ALB; more abdominal pain

Chi-square (cluster vs survival): p = 0.2546 (not significant at = 0.05). Cramér's V = 0.048 (negligible effect). PCA visualization captures only 30.8% of total variance.

11. Model Justification Summaries

11.1 Supervised model comparison (Section 7.5)

Random Forest, Logistic Regression, SVM RBF, and Gradient Boosting were evaluated on the same 702/176 stratified split with random seed 42. All models achieve AUC below 0.63—below the 0.7 threshold commonly cited for clinical utility. SVM RBF achieves the highest macro F1 and class-0 recall, but Random Forest was retained because Gini feature importance is required for the Section 8 synthesis bridge analysis.

Model	Test Acc.	Macro F1	AUC	Importance	Notes
Random Forest (f1_macro tuned)	61.9%	0.550	0.563	Yes (Gini)	85.8% train — reduced overfit vs legacy
Logistic Regression	55.1%	0.539	0.599	No (needs SHAP)	Highest class-0 recall (61.8%)
SVM RBF	61.9%	0.590	0.622	No (needs SHAP)	Best macro F1 & AUC
Gradient Boosting	63.6%	0.482	0.581	Yes (Gini)	High train acc — overfits

11.2 Unsupervised method comparison (Section 6, top)

K-Means, GMM, Agglomerative Ward, and DBSCAN were compared on the same scaled 878×20 matrix. No partition method achieves statistically significant cluster–survival association at k=3. K-Means was selected for highest silhouette at k=3, complete patient assignment (100%), interpretable centroids, and balanced cluster sizes suitable for clinical profiling.

Method (k=3 unless noted)	Silhouette	Chi-sq p	Assignment	Centroids
K-Means	Highest silhouette	$p=0.2546$	100% assigned	Yes — centroids
GMM	Lower silhouette	$p \approx 0.53$	100% assigned	Yes — means
Agglomerative Ward	Mid silhouette	$p \approx 0.65$	100% assigned	No centroids
DBSCAN (best sig.)	Outlier pockets	$p < 0.01$	~49% assigned	No — noise labels

11.3 Why Random Forest? (five factors)

1. Non-linear interactions: all univariate $|r| < 0.15$; trees capture feature combinations.

2. Built-in Gini importance: required for Section 8 synthesis without SHAP/permutation methods.

3. Multicollinearity robustness: random feature subsampling at each split mitigates VIF issues.

4. No scaling required: clean separation from K-Means preprocessing pipeline.

5. Course-appropriate complexity: transparent ensemble mechanics suitable for defense.

12. Compliance, Limitations, and Reproducibility

12.1 Critical rules compliance

Rule	Implementation evidence	Status
No supervised leakage via scaler	StandardScaler fit on full X for K-Means only; RF unscaled	Met
K-Means excludes target	fit_predict on X_scaled; survival overlaid post-hoc	Met
Drop unusable columns	ID, Predict label 1, Predict label 0 removed	Met
Beyond accuracy	F1, macro F1, AUC, CM, classification report, overfitting table	Met
class_weight balanced	All RandomForestClassifier instances	Met
random_state = 42	Split, models, CV, PCA, K-Means	Met
Model justification with metrics	Sections 6.0 and 7.0 + enhanced 7.5 comparisons	Met
Normality testing	D'Agostino-Pearson on 18 continuous features	Met

12.2 Known limitations

1. Moderate class imbalance (2.2:1) despite balanced weights; low class-0 recall persists.

2. Weak univariate correlations (all $|r| < 0.15$) — prediction requires multivariate interactions.

3. Primary RF overfitting persists: 85.8% train vs 61.9% test (23.8 pp gap) after f1_macro tuning.

4. Primary AUC 0.563 — limited discrimination; no model in comparison exceeds AUC 0.63.

5. Cluster–survival association not significant ($p = 0.2546$; Cramér's V ≈ 0.048).

6. K-Means assumes spherical, equal-variance clusters; geometry may differ in reality.

7. Multicollinearity among NLR, PLR, SII, CRP/ALB and component labs.

8. Single internal test split; no external hospital validation cohort.

9. Released features are pre-standardized (z-scores); raw clinical units not in file.

10. Retrospective data — association, not causation; missing staging and treatment variables.

12.3 Reproducibility

1. pip install -r requirements.txt (add reportlab for PDF generation)
2. Open terminal in project root (contains Dataset/ and figures/)
3. jupyter notebook open PCSPF_Data_Science_Capstone.ipynb
4. Kernel Restart & Run All (~3–8 min including grid search and model comparisons)
5. Regenerate this PDF: python scripts/generate_audit_documentation.py

End of document. This record reflects the executed capstone pipeline including all audit enhancements and model justification optimizations. Intended for audit, written report support, and oral defense preparation.